Second moments and conditional prefix exclusion

Source-only derivation; producer-visible written critiques

Exact second moment ⟶ h3 quartic: at most 4 coefficient roots per pair

Sharp bounds contain holes; moment pairs do not uniquely identify support.

Do not enumerate the moment grid expecting a gain

Charge root finding, full collision lists, outputs and complete direct/MITM baselines.

Fix S0 ⟶ at most h lifts ⟶ at most 2h arcs ⟶ test an E2 prefix

O(h) endpoint operations; reject only if every arc misses

No enumeration; semantic checker required; no frequency or runtime claim.

Exact result and necessary-bound controls

For h3, fixing the support coefficients u,j gives $q^4+(2u-6a)q^2+8bq+u^2-2au-3a^2-4j=0$, hence at most four distinct field coefficient candidates. Certified complete root extraction and all remaining rows/witnesses are still required. At h4, q=0 must retain the full A1 fiber whenever its forced moment is allowed.

For offsets d_i between 0 and M=B−1, write S=sum d_i and E2=sum of pair products. Then $j=\binom{h}{2}t^2+(h-1)tS+E2$ and $2E2=S^2-\sum d_i^2$. Balanced offsets minimize squares; packed offsets maximize them, giving sharp conditional extrema. Interior holes remain: B4,h3,S3 permits E2=0,2,3 but not1. Multisets (0,3,3) and (1,1,4) share S6,E2=9, so the moments do not determine support.

The moment-coordinate change is affine; further coefficient elimination need not be. No regularity reduction is inferred. Saturation disables the E2 bound for that sum, not all trace or other-sum pruning.

Cheap prefix test instead of a moment grid

Fix full canonical S0. There are at most h integer lifts in the permitted sum interval and at most 2h canonical E2 arcs [L,U]. For an LSB prefix r at depth d, H=2^d, test $\lceil (L-r)/H \rceil \le \lfloor (U-r)/H \rfloor$. Reject only if every arc misses; signed floor/ceiling are mathematical, not truncation toward zero. Here ell=ceil(log2 p) and $0 \le d \le ell$. This uses O(h) integer operations on O(log p)-bit operands, plus authentication/parsing and O(ell) input/clause work; no moment or coefficient enumeration is needed.

The clause fixes all S0 bits and the E2 prefix. E2 may remain auxiliary, preserving original q projections. A new authenticated semantic lemma or checked CNF derivation is necessary: current prefix-only/DRAT-RUP verification does not automatically accept it. Passing the test does not certify an actual support. No rejection-frequency or runtime claim is made.

Baselines, scope and next action

The moment-pair envelope is $\Theta(B^3)$ when p grows faster than B^2. Complete h3/h4/h5 baselines have $O(B^2+N)$, $O(B^2 \log(B+1)+N)$, $O(B^3 \log(B+1)+N)$ operation envelopes, plus preparation and all arithmetic/witness costs. Joins retain every collision, repeated factor and identity intermediate sum. Do not scale a naive moment grid on the bounds alone; specify the conditional-prefix proof interface next.

Original six coefficient cells are coset-based, not interval charts. Old implementation prohibitions and all106+6/five-lead budgets remain. No imports, code, trials, retries or IDs occurred. Reviews are non-blinded critiques; omitted initial plan fields and absent blind rederivation are explicitly recorded, not a native gate pass.

Sources: frozen design-v1.md, baseline-v1.md, controls-addendum-v1.md, prefix-arcs-addendum-v1.md and their critiques; source-bindings.json; definition-clarification.md; review-procedure-supplement.json. Prior art: Petit–Kosters–Messeng, prime-field algebraic approaches Wagner2002, ordinary joins. No novelty or transplanted filtered-XOR speedup claim.